

31338 Network Servers

32520 Systems Administration

Week 3

Networking and Services

(Networking and miscellaneous networked services)

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7 Network Layers

- **The Physical Layer #1**
 - Hardware, interface card
- **The Data Link layer #2**
 - Hardware/interface card Device driver
- **The Network layer #3**
 - IP (Internet Protocol) (data packets)
 - ICMP (Internet Control Message Protocol) (ping, error messages).
- **The Transport layer #4**
 - TCP (Transmission Control Protocol) - error correcting connections (e.g. email)
 - UDP (User Datagram Protocol) - no order, no guarantee (e.g. sound player)
- **The session layer #5 and the presentation layer #6**
- **The Application layer #7**
 - FTP, Telnet, rlogin, ssh, SMTP...

Network Addresses

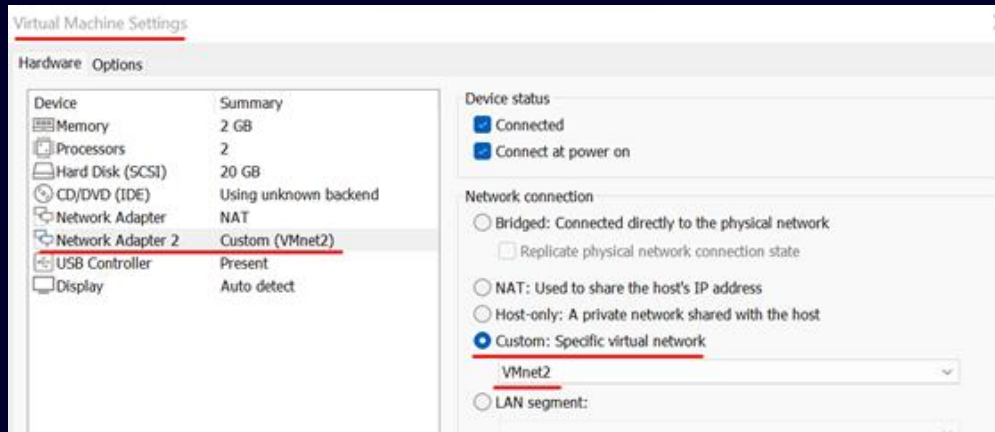
- **The Data Link layer** - MAC Addresses
 - MAC = Media Access Control
 - Using `ifconfig`: `00:A0:CC:24:BA:02`
 - First half - Company ID : Second half - Product ID
- **The Network layer**
 - 4 bytes IP Addresses: Network portion (NP) and Host portion (HP) defined by Network Mask. For example, given an IP address: 192.168.3.3
 - Network Mask defines the 1st part as the NP and 2nd part as HP 255.255.255.0
 - » Network portion 24 bits = 11111111 11111111 11111111
 - » Host portion 8 bits = 00000000
 - » 192.168.3.3/24 (NP)
 - 16 bytes or 8 16-bit segments IP v6 Addresses (128 bits) – larger address space for unicast, multicast and broadcast
 - Broadcast -- send a message or packet to all devices on a network
 - All computers on the network
 - Logically: IP = 192.168.255.255
 - Testing broadcast: `ping -b 192.168.255.255`

Network Addresses

- Look at the IPv4: 4 Bytes/32bits!
- Private IP addresses range:
 - **10.0.0.0 to 10.255.255.255 (10.0.0.0/8), 172.16.0.0 to 172.31.255.255 (172.16.0.0/12), 192.168.0.0 to 192.168.255.255 (192.168.0.0/16)**
 - Can be freely used in private networks
 - **Cannot be routed over the Internet (excluded from public access fileting)**
- Public IP addresses
 - IPs allocated by a central authority IANA Internet Assigned Numbers Authority
 - Authority delegated to regional centres (e.g. to **ISP**)
 - Does not include private IPs and a few others e.g. link local
- Network Address Translation (NAT)

NAT translates private IP addresses in an internal network to a public IP address before packets are sent to an external network.

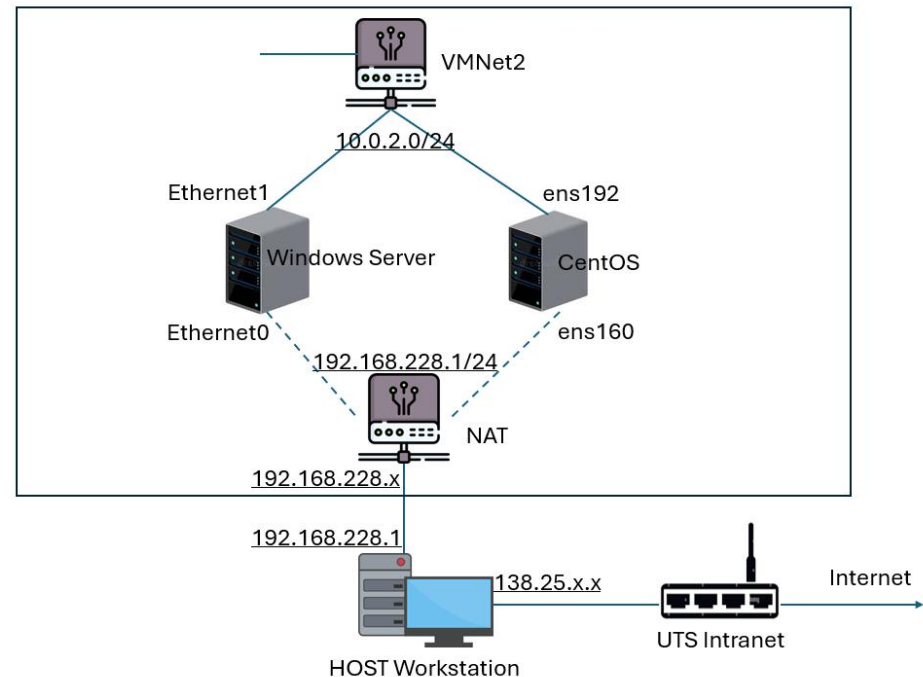
Network Addresses



Two virtual interface (ens160 and ens192) or called v-NIC (Network Interface Card)

For vmnet2, private network portion:
10.0.2.0/24 – 10.0.2.254/24.

For NAT, public network portion:
192.168.228.0/24 –
192.168.228.254/24



Network Configuration in Lab 3

- Transient setting for ad hoc settings (legacy)
 - **ifconfig**
 - `ifconfig ens192` #IP address, Network mask, MAC address, etc
 - `ifconfig ens192 192.168.168.64 netmask 255.255.255.0` #setting
 - `ifconfig ens192 hw ether xx: xx: xx: xx: xx: xx` # configure MAC
 - **route**
 - `route -n` #show current routes
 - `route del default gw xxx.xxx.xxx.xxx` #delete default gateway
 - `route add default gw xxx.xxx.xxx.xxx` #add default gateway
 - `route add -net 192.168.1.0 netmask 255.255.255.0` # add an entry to the routing table.
 - `hostname domainname` #hostname.domain, e.g., email.itd.uts.edu.au

Network Configuration in Lab 3

- Transient setting for **ad hoc** settings: *ip* utility
 - check, set IP address (`man ip; ip addr help`)
 - state is up unless manually put to down
 - `ip a; ip addr; ip addr show; ip address show; ip -4 addr; ip addr show dev ens160 #show ip address`
 - `ip addr add 192.168.0.10/24 dev ens160 # {add | change | replace},`
support multiple address
 - link down/up
 - `ip link show; ip link show ens160`
 - `ip link set ens160 down; ip link set ens160 up`
 - route, default gateway
 - `ip route`
 - `ip route add 192.168.0.10/24 dev ens160 metric 100`
 - `ip route delete 192.168.0.10/24 dev ens160`

Network Configuration in Lab 3

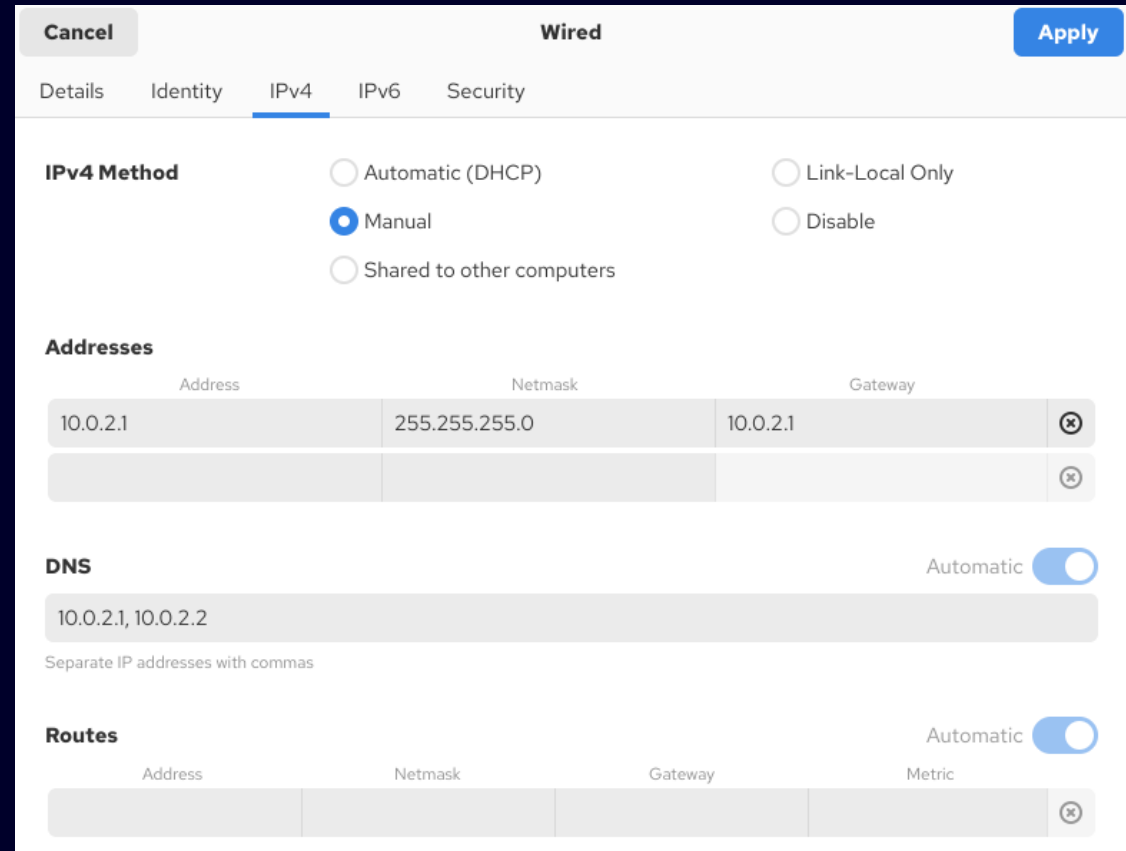
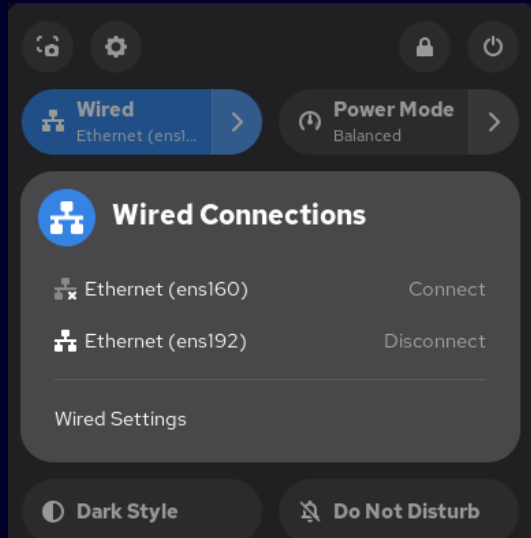
- **NetworkManager (NM):** default networking service
 - In CentOS 10, change network status by using
`systemctl {is-active|restart|start|enable} NetworkManager`
 - nmcli is a command-line tool which is used for **controlling** NetworkManager. It can also be used to display network device status, create, edit, activate/deactivate, even delete network connections. Then run `nmcli c <adaptor_name> up` or/and `nmcli c <adaptor_name> down`.
 - **NM functions:** Keep network device and connection up and active when they are available: create **temporary connections** to provide connectivity even without network configuration file in the system for reboot.
 - **NM Tools:** 1) GUI (`nm-connection-editor`, GNOME), `nmtui`(NetworkManager's Text User Interface) and `nmcli` (**often used!**)
 - **NM Support:** scan interface (e.g., WiFi), manage IP address, static route, DNS information, VPN connection, etc.
 - **NM also Support:** configuration files
 - **If stop NM,** The current network setup is still available. But if you restart the computer, it won't connect the network! So, you should use the option `enable` in the `systemctl` command to automatically activate it when the system reboots.

Network Configuration

- Persistent change on networking configuration files:
 - `/etc/NetworkManager/system-connections/<adaptor_name>.nmconnection` **Centos 10**
 - interface-specific configuration for `ens160` and `ens192`
 - id (e.g., `ens192`), `autoconnect` (true or false), interface-name (e.g., `ens192`)
method (manual or auto), `address1`, `dns`, `dns-search`
 - Require manually put interface down and up: `nmcli c down ens192`
 - `/etc/resolv.conf`
 - define DNS server, updated automatically **to resolve the mapping between domain name and IP address.**
 - **domain** *it.uts.edu.au* # Domain name to be used locally
 - **search** *it.uts.edu.au* # Search list to be used for hostname lookup
 - **nameserver** *8.8.8.8* #IP address of the name server that the resolver will use.

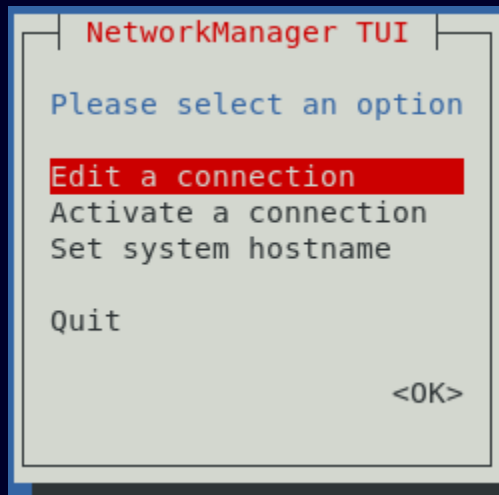
Network Configuration (GUI)

- Persistent change through **settings for CentOS**:



Network Configuration (text-menu)

- Persistent change through tools:
 - **nmtui**: text-based menu tool
 - All updates will be written to ifcfg file. Activate means “ONBOOT=yes”
 - **nmtui** #Tab, Shift+Tab, Enter, Space



Network Configuration (Cmd-line)

- Temporary change: **nmcli: text-only command-line tool**

- The **nmcli** command uses the following **syntax**:

nmcli [OPTIONS] OBJECT [COMMAND] [ARGUMENTS...]

nmcli # controlling NetworkManager and reporting network status check ip.

- The **OBJECT** field can be one of the following:

Object	Meaning
general	Display information about or modify the status of NetworkManager.
networking	Display information about or modify the network managed by NetworkManager.
connection	Display information about or modify connections managed by NetworkManager.
device	Display information about or modify devices managed by NetworkManager.
radio	Display the status of, and enable or disable, the radio switches.

-t display output
-p easily readable

Network Configuration

Connection: Display information about or modify connections managed by NetworkManager.

- **connection:active (colored)/inactive**
 - `nmcli c modify ens192 ipv4.addr '192.168.1.200'`
 - `nmcli c up ens192` #activate connection
 - `nmcli c down ens192` #de-activate connection
 - `nmcli c delete ens192` #delete connection
 - `nmcli c show` #check connection list, name, uuid (universally unique identifier), type, device
 - `nmcli c show ens160 | grep address` #show more details

Network Configuration

Device: Display information about or modify devices managed by NetworkManager.

- **device:** connected/disconnected, unmanaged, unavailable
 - Connected: managed by NM, and active connection
 - Disconnected: managed by NM, no active connection
 - Unmanaged: not managed by NM
 - Unavailable: not available, cannot be managed by NM
- `nmcli dev | nmcli d` #check device list
- `nmcli d status` #show device status
- `nmcli d show ens160` #detailed information
- `nmcli d connect ens160` #activate NIC, and refresh active connection, if no active connection, will automatically select one and active it
- `nmcli d disconnect ens160` #disconnected, de-activate connection

Network Configuration

- Recommendation
 - distinguish transient change and persistent change
 - modify network configuration file: `nmconnection` file and bring interface up: `nmcli con up <adaptor_name>`
- Change `nmconnection` file directly (**highly recommended!**), make sure you understand this configuration file.
 1. `vim /etc/NetworkManager/system-connections/ens192.nmconnection`
 2. `nmcli c reload ens192` #enable ens37 effective immediately
 3. `nmcli c up ens192` #bring interface up

Hostname

- Three classes of hostname: **static**, **pretty**, and **transient**
 - **static**: traditional hostname, stored in `/etc/hostname` file
 - **pretty**: free-form **UTF-8*** hostname for presentation to the user
 - **transient**: dynamic hostname maintained by the kernel: localhost by default, can be changed by DHCP or multicast DNS at runtime (**temporal change!**)
- Example of transient change: `hostname hostname.example.com`
- **Persistent change** saved in `/etc/hostname`
 - Manual change in `/etc/hostname` will not be effective automatically, must run `systemctl restart NetworkManager`, or reboot, or `systemctl restart systemd-hostnamed` to make it effective
- **hostnamectl** (static/transient/pretty)
 - `hostnamectl status` `#query`
 - `hostnamectl set-hostname <host_name>` `#effective instantly`

* UTF-8 is an encoding system for Unicode. It can translate any Unicode character to a matching unique binary string and can also translate the binary string back to a Unicode character. UTF = "Unicode Transformation Format"

DNS & DNS-search

- Steps when only one NIC (Network Interface Controller) enabled

1. `vim /etc/NetworkManager/system-connections/ens192.nmconnection`

2. `nmcli c reload ens192 #reload`

3. `nmcli c up ens192 #activate`

- Sample `ens192.nmconnection`

```
[connection]
id=ens192
uuid=da330b4e-d39c-37c4-9d35-00ab32239557
type=ethernet
autoconnect=true
interface-name=ens192

[ipv4]
address1=10.0.2.1/24
method=manual
gateway=10.0.2.1
dns=8.8.8.8
dns-search=it.uts.edu.au;

dns-priority=10
```

`/etc/resolv.conf`

```
# Generated by NetworkManager
search it.uts.edu.au localdomain
nameserver 8.8.8.8
nameserver 192.168.228.2
```

DHCP: Dynamic Host Configuration Protocol (Lab 4)
DNS: Domain Name System (Lab 6)

Iptables

- Provide a firewall (IPv4 and IPv6) between the LAN and Internet packet filter rules in the Linux kernel.

***iptables** is a user-space utility program that allows a system administrator to configure the IP packet filter rules of the Linux kernel firewall,

- IP **masquerading** (NAT) – A bad thing !
 - Rewrites IP address in header to enable Internet access from private IPs
- Packet filtering
 - list of rules, checked in order
 - first match usually terminates search through list
 - unmatched packets will follow default policy
 - can match on source/destination IP address, source/destination port, protocol, state, etc.

- **firewalld** VS. **iptables**: firewall can filter requests based on protocol or target-based rules. Both **firewalld** and **iptables** are the tools for managing firewall rules on a Linux machine.
 - **firewalld** acts as a front-end tool via the **iptables** back-end utility
 - **firewalld** can update rules when process is still running
 - **firewalld** can set up a default zone that drops all incoming and outgoing traffic and then selectively allows specific traffic,
 - By default, **iptables** blocks all incoming traffic and allows all outgoing traffic.
 - **firewalld** Define 9 zones: block, work, home, public, trusted, drop, DMZ, internal, external. /usr/lib/firewalld/zones
 - Tools: **firewall-cmd**, **firewall-config** (GUI)
- **firewall-cmd**
 - `yum install firewalld firewall-config`
 - `systemctl start|stop|restart|enable|disable firewalld`
 - `firewalld-cmd --state` #check status
 - `firewall-cmd --zone=public --add-port=80/tcp --permanent`

System Time

- **Unix Time*** = GMT (Greenwich Mean Time)
 - Also called UTC (Universal Time Coordinated)

***Unix time is a date and time representation widely used in computing. It measures time by the number of non-leap seconds that have elapsed since 00:00:00 UTC on 1st January 1970, the Unix epoch**
- **date** with no options to display current time
 - Sat Aug 16 01:07:13 AEST 202
- **date -u** # print UTC time
- Abbreviations (UTC, GMT, EST...)
[<http://timeanddate.com/library/abbreviations/timezones/>]
- Changing local timezone (remove first)
 - **ln -s /usr/share/zoneinfo/Australia/Sydney /etc/localtime** **(ls -l symbolic link)**

- Change time (legacy)
 - Change system time – **date**
 - **date +%A, %B %d %Y** # define format
 - **date MMDDhhmm[[CC]YY][.ss]** #month, date, hour, minute; century, year, seconds
 - Date 12101430 #set to 10 Dec. 14:30
 - Change hardware time – **hwclock**
- **timedatectl** tool: set time/zone/system/hardware
 - **timedatectl** #print
 - **timedatectl set-time "2021-08-20 06:15:00"**
 - **timedatectl set-ntp 0** #de-active
 - **timedatectl set-ntp yes** #start chronyd daemon
- What is NTP? The Network Time Protocol (NTP) is used by hundreds of millions of computers and devices to synchronize their clocks over the Internet. If your computer sets its own clock, it likely uses NTP.
- Chronyd is a daemon for synchronisation of the system clock with NTP

NTP Time

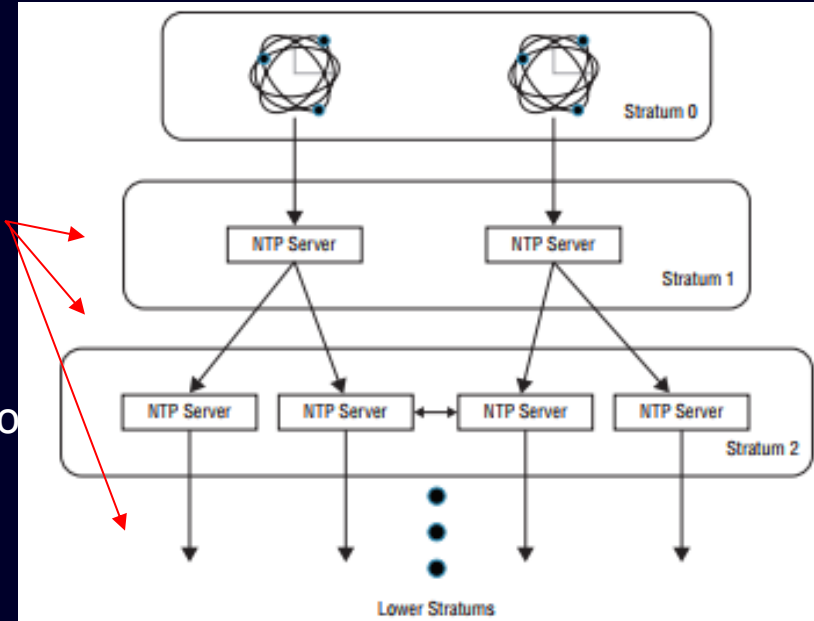
- Problem: computers forget the time
- Network Time Protocol: synchronize clocks over networks
 - Client-server mode
 - Clock **stratum** scheme: layered from most accurate **0-15**
 - Allow packets carry time data to traverse more quickly between NTP server and client
 - Programs that implement NTP will adjust time in order to offset NTP packet travel time

- Configure server side:

- Install `yum install ntp`
- Keep the time in sync
 - `systemctl start ntpd`

- Synchronise client side

- Correct the time now:
 - `ntpdate time.uts.edu.au`



- **chrony** daemon
 - Faster and Keep accurate time even on busy network or down for some time.
 - Configure: `/etc/chrony.conf`
 - `systemctl start chronyd`
 - `chronyc sources -v` #check sources
 - `chronyc sourcestates` #time server stats
 - `chronyc add server time.uts.edu.au` #add NTP server
 - `chronyc tracking` #view software clock information
 - `chronyc makestep` #only sync system date and time

Chronyd is a daemon for synchronisation of the system clock with NTP